

trajectory-judge: What Outcome-Only LLM Judges Miss on Agent Trajectories

Hadi Mohammadi

Department of Methodology, Statistics & Data Science
Utrecht University, The Netherlands
h.mohammadi@uu.nl

 github.com/mohammadi-hadi/trajectory-judge

 `pip install trajectory-judge==0.2.0`  `doi:10.5281/zenodo.21797926`

Abstract

A direct test of an LLM judge of agent trajectories injects faults into correct runs and reports recall, per fault type or by whether the fault broke the environment outcome (*loud*) or not (*silent*). Such recall can credit a judge with detection it does not have; *paired discrimination*, its flag rate on the faults minus its rate on the clean runs they came from, exposes this. Our testbed, a deterministic support desk with a scripted oracle and a one-step fault injector, labels all 400 trajectories exactly. A 14B judge shown only the request and final reply scores 34% to 76% recall on four fault types that leave the reply unchanged. There its input is the clean run’s, so its paired discrimination is zero and that recall is its flag rate on clean runs. Splitting by outcome survival does not fix this: its loud recall of 84% is a paired +0.393 and its silent recall of 45% a paired +0.048, all from the two fault types that change the reply. Told to check each step, the same model flags every fault of those four types and 0 of 100 clean runs (95% CI up to 3.6%). It does not reliably check the reply: of four invented promises it flags one every time and the other three once in 42 faults. Shown every step but asked only about the reply, it still reaches a paired +0.69 on reply-unchanged faults, against +1.00 when told to check each step. We recommend reporting paired discrimination against clean parents, split by whether the fault reaches the judge’s input and by outcome survival, and release the testbed, raw verdicts and analysis pipeline.

1 Introduction

Agents that call tools are often gated on outcome-level judgment: a model reads the user’s request and the agent’s final reply and decides whether the case was handled well. Some faults never reach the reply. A refund of the correct amount issued without the required eligibility check produces the same reply as one issued correctly (Figure 1). A direct test injects faults into correct runs and reports recall, per fault type or by whether the fault broke the environment outcome (*loud*) or not (*silent*). We show that such recall can credit a judge with detection it does not have, and that pairing each injected fault with the clean run it came from, while labelling whether the fault reaches each judge’s input, exposes this.

That a judge shown only the request and the reply cannot see a fault absent from the reply is expected, and serves here as a control with a known answer. What pairing measures is how far recall overstates detection. Recall inherits the judge’s base rate in each scenario: our injector leaves the reply byte-identical to the clean run’s on four of six fault types, and a 14B judge shown only the reply scores 34% to 76% recall on them, yet its paired discrimination Δ there (flag rate on the faults minus flag

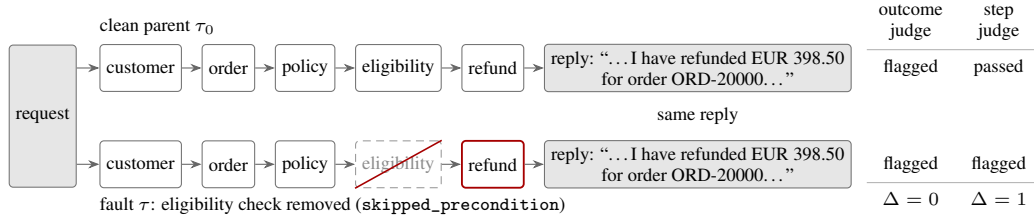


Figure 1: A fault and its clean parent. Shaded: what the outcome judge reads. The reply does not change, so it flags both runs; the step judge flags only the fault.

rate on their clean parents) is zero. That recall is its flag rate on clean runs, 0% to 76% by scenario; on a newer inference engine the same prompt scores 20% to 26% there, again with no discrimination (§7.2). Splitting recall by outcome survival is not sufficient. Of the judge’s 105 flags on loud faults, 55 are reply-unchanged faults flagged as often as their parents, and after pairing its loud recall of 84% is +0.393, all of it from the one loud type that changes the reply. The reply itself is discussed but rarely checked: told to check each step, the same model’s paired discrimination on an invented promise (+0.18) is close to the reply-only judge’s (+0.16), mostly from one of the four invented sentences. The two judges also differ in instruction; an ablation crosses it with view (§7.2).

Evaluating an evaluator usually needs human labels, which puts the labeller under test too. We avoid this by construction: a deterministic support-desk environment, a scripted oracle that is correct by construction and checked by a test on every generated instance, and a fault injector that makes one edit to an oracle run at a known step and replays the edited calls, so every observation comes from the environment. Each trajectory carries an exact label at no annotation cost: faulty or not, step, type, whether the environment outcome survived, whether the reply changed, and its clean parent. With the agent fixed, an error belongs to the judge or to its view, not to the agent or to annotators.

Contributions. (i) A measurement design for trajectory judges: each injected fault is paired with its clean parent, the judge’s own control, and labelled by whether it reaches each judge’s input and whether the environment outcome survived (§3, §5). (ii) A testbed that implements it: a permissive environment with a strict checker, a correct-by-construction oracle, a six-type single-fault injector, and a rule baseline whose coverage is pinned by tests (§3). (iii) Findings scoped to this testbed and two local models: on reply-unchanged faults a reply-only judge’s recall is its base rate; outcome-stratified recall needs pairing; a judge told to check each step flags every reply-unchanged fault and 0 of 100 clean runs; no judge reliably verifies the reply; majority voting over three samples adds cost without a measurable gain; an 8B judge does not discriminate under our prompt (§6, §7). (iv) An artifact that rebuilds every number in this paper offline from the released raw verdicts (§10).

2 Related work

Reliability of LLM judges. Model-as-judge evaluation is standard for open-ended outputs [Zheng et al., 2023, Liu et al., 2023, Gu et al., 2026], and its pathologies are well documented: position bias [Wang et al., 2024a], verbosity bias [Zheng et al., 2023], self-preference [Panickssery et al., 2024], objective failures on instruction-following comparisons [Zeng et al., 2024], and leniency and imperfect agreement with human scores even on a simple task [Thakur et al., 2025]. That line audits judges of single responses against human labels; we audit judges of multi-step tool trajectories against labels correct by construction, so a flag on a clean trajectory is a false positive. The judge design itself (reasoning before the verdict) follows Mohammadi et al. [2026], and treating a model’s labels as measurements that need their own reliability estimate follows Mohammadi et al. [2025b]; Mohammadi [2026] develops both at thesis length and notes that a model’s stated reasoning may not reflect how it decided [see also Turpin et al., 2023, Mohammadi et al., 2025a].

Agent benchmarks. Benchmarks for tool-using agents score the agent, increasingly on the trajectory rather than the outcome alone: τ -bench and its successor compare final database state [Yao et al., 2025, Barres et al., 2026], AgentBench and WebArena score task success [Liu et al., 2024, Zhou et al., 2024], AgentBoard tracks subgoal progress [Ma et al., 2024], and recent work scores reasoning and tool-use trajectories directly [Kim et al., 2026, He et al., 2026]. We invert the roles: the agent is a fixed oracle, and the evaluator is the system under test.

Monitoring and error detection. AI-control monitors score actions or transcripts for covert sabotage [Bhatt et al., 2025, Kutasov et al., 2025], and step-level guardrails flag unsafe tool calls [TS-Bench; Mou et al., 2026]. Judges of agent trajectories are checked against benchmark outcomes or human labels [Pan et al., 2024, Lù et al., 2025, Fan et al., 2026], and an agentic judge inspecting intermediate work outperforms an LLM judge [Zhuge et al., 2025]. What the evaluator sees matters: chain-of-thought monitors can far outperform monitors of actions and outputs [Baker et al., 2025], state-based success checks credit τ -bench runs that violate the procedure or return nothing [Cao et al., 2026, Zhu et al., 2025], and LLM judges struggle to catch false completion claims [Advani, 2026].

Failure attribution and fault injection. Locating the failing step or agent in a trace is an emerging task [Zhang et al., 2025a, Deshpande et al., 2025, Cemri et al., 2025, Ma et al., 2025]. Injected faults supply labels for failure attribution [Zhang et al., 2026] and for judges of dialogue compliance [Yang et al., 2026] and of research-agent traces [Wang et al., 2026]. Paired perturbations expose evaluator blind spots on single responses [Doddapaneni et al., 2024], and concurrent work tests localisation on view-identical trace pairs [Zhu and Pu, 2026]. What we add is how each fault and its clean parent are used: the parent is the judge’s own control, giving paired discrimination, and the fault is labelled by whether it reaches each judge’s input, the propagation condition that separates weak from strong mutation [Howden, 1982]. Without the pairing, per-type and outcome-stratified recall can credit a judge with detection it lacks (§7.1). Mutation testing has seeded known defects to measure a detector for half a century [DeMillo et al., 1978, Jia and Harman, 2011]. Whether mutants substitute for real faults is the classic validity question [Just et al., 2014]; our paired results cover injected faults, whose mix is unlike an agent’s own (§8).

Process versus outcome supervision. In mathematics, step-level feedback trains better verifiers than outcome-level feedback [Lightman et al., 2024, Wang et al., 2024b] and is needed for correct reasoning steps even when final-answer error is similar [Uesato et al., 2022]; ProcessBench measures error localisation in reasoning chains [Zheng et al., 2025], on which recent process reward models are evaluated [Zhang et al., 2025b]. That agenda trains scorers where the final answer is checkable; we ask the measurement question underneath it, whether an off-the-shelf judge can see process faults at all when they leave the final reply unchanged [cf. Lang et al., 2024], and find that one reading only the request and reply cannot, while the same model told to check each step can.

Calibration. We score each judge’s confidence with expected calibration error [Pakdaman Naeini et al., 2015, Guo et al., 2017] and the Brier score. Confidence read from token probabilities [Kadavath et al., 2022] or verbalised [Tian et al., 2023] can be usable, but verbalised confidence trends overconfident [Xiong et al., 2024].

3 A testbed with ground truth by construction

Environment. The environment is a customer-support desk with seven tools (`get_customer`, `lookup_order`, `get_policy`, `check_eligibility`, `issue_refund`, `escalate`, `reply`) and a written standard operating procedure: verify the customer, look up the order, read the policy for the item, confirm eligibility before moving money, refund exactly the authorised amount, escalate when not eligible, and reply asserting only what the observations support. Instances are generated in six scenarios: full-price refunds, restocking fees, expired windows, non-refundable items, orders belonging to a different customer, and orders already refunded, drawn round-robin from a single seeded stream (Appendix A). The environment is permissive and the checker is strict: `issue_refund` will refund an order whose eligibility was never checked, exactly as a real payments API would. Nothing in the world stops an agent from skipping the process; only the rule checker says it was wrong. Without this there would be no silent faults to measure, so a test (`test_environment_is_permissive_by_design`) pins it.

Notation. A trajectory $\tau = (g, s_0, \dots, s_{T-1}, a)$ pairs a goal g and final reply a with steps $s_t = (h_t, c_t, o_t)$: a thought, a tool call $c_t = (\text{tool}_t, \text{args}_t)$, and an observation o_t . Its label $\ell(\tau) = (y, t^*, \phi, \omega)$ records faulty or not, the failure step, the type $\phi \in \Phi$ ($|\Phi| = 6$), and the *environment outcome* ω : whether the episode ends in the refund amount or escalation the instance requires, with reply text excluded. A fault is *silent* iff $y \wedge \omega$ and *loud* iff $y \wedge \neg\omega$, and every faulty τ has a *clean parent* τ_0 , the oracle run it was injected into. A judge maps a *view* of τ to a verdict $(\hat{y}, \hat{t}, \hat{\phi}, \hat{c}, r)$ with stated confidence $\hat{c}$ and rationale r . The step view $V_{\text{step}}(\tau)$ is the full rendering. The outcome-only view $V_{\text{out}}(\tau) = (g, a)$ follows LLM-as-judge usage [Zheng et al., 2023] and holds the request and

Table 1: The six fault types, 50 faults each.

Fault type	The single edit	Rule recall	Env. outcome survives	Reply changed
wrong_tool	order fetched again in place of the policy	0.00	always	no
hallucinated_argument	policy fetched for an invented SKU	1.00	always	no
skipped_precondition	refund without the eligibility check	1.00	at full price	no
ignored_observation	refund of an unauthorised amount	1.00	never	no
premature_stop	stops before acting or replying	1.00	never	yes
unsupported_claim	invented promise added to the reply	0.00	always	yes

the reply, never the end state. We call the judge on this view the *outcome judge*. Unlike a state check in the style of τ -bench [Yao et al., 2025], which flags every loud fault, it cannot see a loud fault that the reply does not reveal. Every fault changes V_{step} , and $V_{\text{out}}(\tau) = V_{\text{out}}(\tau_0)$ iff the fault leaves the reply unchanged. For any judge whose verdict depends only on (g, a) , such a fault has its parent’s verdict distribution, so its expected paired discrimination (§5) is zero; under greedy decoding in the August runs (§4) we observe identical verdicts, confidences and rationales on 151 of 151 pairs. This does not cover judges given tools, retrieval or the environment state, and the share of faults it covers (200 of 300) is set by our injector.

Oracle. The oracle is a fixed six-step script, identical across scenarios up to one branch: verify, look up, read the policy for the SKU returned by the lookup observation, check eligibility, then refund the authorised amount if eligible and escalate otherwise, and reply. A test parametrised over all instances asserts it violates no rule and reaches the expected outcome.

Fault injector. Each mutation edits the oracle’s call list at one step and replays the edited list against a fresh environment, so a hallucinated SKU fails its lookup and no observation is written by hand; a test asserts that the replayed observations are consistent with the calls. Each mutation targets one failure: *skipped_precondition*, for example, refunds the order total so that the amount stays grounded in the lookup and no second checker rule fires. Four types leave the reply unchanged (200 faults: 125 silent, 75 loud); *premature_stop* replaces it with a holding message and *unsupported_claim* appends a sentence (Table 1). Keeping the oracle’s reply has one side effect a real run would not have. In the 75 loud faults (all of *ignored_observation*, and *skipped_precondition* under a restocking fee) the reply states the authorised amount although the order total was refunded. The agent of §8 states the amount it refunded in 20 of 20 refund replies, and 0 of its 60 replies equal the oracle’s. The stale amount is invisible to a judge that reads only the reply, but it gives one that reads the steps a second cue. Injection is string-seeded per (instance, type); the judged set is 400 trajectories (100 clean, 175 silent, 125 loud) in a fixed shuffled order.

The rule engine and its two zeros. A checker walks each trajectory once, keeping a pool of grounded values seeded from the goal, and enforces eight rules over grounding, ordering, amounts, identity, and terminality (Appendix B). It catches every fault of four types and none of the other two. These zeros are limits of the eight rules: none encodes the required tool order, so a grounded but redundant *lookup_order* in place of *get_policy* breaks nothing, and none reads the reply, since prose is excluded from the grounding pool. A required-sequence rule, written post hoc and kept out of the baseline, catches 50 of 50 *wrong_tool* faults with 0 false alarms on clean runs, but also flags 27 of the 47 agent episodes the checker passes (Appendix B). The rule engine is thus a fixed reference point that does not bound programmatic checking; a test fails if either zero changes.

4 Five judges

Table 2 lists the five judges [Yang et al., 2024, Grattafiori et al., 2024]. The LLM judges share the procedure and taxonomy word for word and decode under a JSON schema with reasoning first. The outcome and step judges differ in view, task instruction and schema (the outcome judge names no step and is told that unseen steps are not evidence of a failure), so comparing them compares configurations; on reply-unchanged faults the view is necessary whatever the instruction (§3), and §7.2 crosses the two. The ensemble [Wang et al., 2023] is a majority over three step-judge samples at $T=0.7$; its confidence, the vote share, takes two values. The LLM judges ran in August 2026 on

Table 2: The five judges (prompts and schemas: Appendix D).

Judge	View	Task	Model	Decoding	Confidence
rules	steps	eight rules	n/a	deterministic	two constants
outcome	(g, a)	judge the reply	qwen2.5:14b	$T=0$, seed 7	stated
step 14B	full	check each step	qwen2.5:14b	$T=0$, seed 7	stated
step 8B	full	check each step	llama3.1:8b	$T=0$, seed 7	stated
selfcons	full	check each step	qwen2.5:14b	$T=0.7$, seeds 7–9	vote share

Table 3: Main results on 400 trajectories. Brackets: 95% intervals; †: exact Clopper–Pearson.

Judge	Recall			False	Paired	Localisation		Type	
	all	silent	loud	alarms	Δ	detected	joint	F1	ECE
rules	0.667	0.429	1.000 [0.97, 1.00]†	0.000 [0.00, 0.04]†	+0.602 [+0.58, +0.63]	1.000	0.667	0.667	0.075
outcome	0.613 [0.57, 0.66]	0.451 [0.38, 0.52]	0.840 [0.78, 0.90]	0.330 [0.24, 0.43]	+0.163 [+0.12, +0.21]	n/a	n/a	0.300 [0.27, 0.33]	0.253
step 14B	0.857 [0.84, 0.88]	0.766 [0.74, 0.80]	0.984 [0.96, 1.00]	0.000 [0.00, 0.04]†	+0.829 [+0.80, +0.86]	0.844 [0.82, 0.87]	0.723 [0.70, 0.75]	0.606 [0.55, 0.65]	0.033
step 8B	0.990 [0.98, 1.00]	0.983 [0.96, 1.00]	1.000 [0.97, 1.00]†	1.000 [0.96, 1.00]†	−0.012 [−0.03, +0.00]	0.747 [0.71, 0.78]	0.740 [0.71, 0.77]	0.311 [0.27, 0.35]	0.148
selfcons	0.843 [0.82, 0.87]	0.760 [0.73, 0.79]	0.960 [0.92, 0.99]	0.010 [0.00, 0.03]	+0.813 [+0.78, +0.84]	0.885 [0.86, 0.91]	0.747 [0.72, 0.77]	0.583 [0.53, 0.63]	0.084

Ollama 0.30.11 (the August runs; code release v0.1.0); the ablation of §7.2 ran in October on 0.33.3. Prompts are verbatim in Appendix D.

5 Metrics and uncertainty

Detection is binary per trajectory. We lead with sensitivity (recall over all, silent and loud faults) and specificity (one minus the false-alarm rate on clean runs), which, unlike precision, F1, Brier score and ECE, do not depend on the share of faults in the set (here 300 of 400; Appendix E). Paired discrimination Δ_J is the mean of $\hat{y}_J(\tau) - \hat{y}_J(\tau_0)$ over faults τ whose parent was judged (251 of 300): the share of faults judge J flags minus the share of their clean parents it flags. Recall mixes detection with the judge’s flag rate in the fault’s scenario; Δ removes the latter. The ablation (§7.2) also judges the other 41 parents. Localisation is exact match on the failure step, over the faults a judge detected (a missing or out-of-range step is a miss) and jointly over all 300 faults (detected and exact). `premature_stop` is labelled at its last executed step, so a judge that names the missing next action scores a miss (Appendix C); we also report the other five types alone (Table 7). Typing is six-class macro-F1 over faulty trajectories and the share of all faults detected and correctly typed; calibration is ten-bin ECE [Pakdaman Naeini et al., 2015] against the binary verdict, coarse for the rule engine’s two constants and the ensemble’s two-valued vote share; cost is seconds per trajectory on one workstation (Brier scores and tokens: Appendix E).

Intervals are 95% percentile bootstrap intervals [Efron, 1979] ($B=10,000$) that resample trajectories within the eight design cells (clean, and each fault type by outcome), one resample shared across judges. The rule engine is constant within a cell, so its entries get no such interval. Pooled paired estimates resample instances, since one parent serves several faults. Recalls and false-alarm rates at 0 or 1 in Table 3 get Clopper–Pearson intervals (†), and single-type paired rows get exact McNemar tests. Intervals are conditional on the generation seed and greedy decoding (Appendix F).

6 Results

Table 3 gives the comparison; Figure 2 sets the two single-pass 14B judges’ flag rate on each fault type against their rate on its clean parents. The outcome judge flags 0.840 [0.78, 0.90] of loud faults and 0.451 [0.38, 0.52] of silent ones, and 0.330 [0.24, 0.43] of clean runs: much of that recall is not detection, and its paired Δ is +0.163 (§7.1). The step judge, the same model told to check every step, reaches

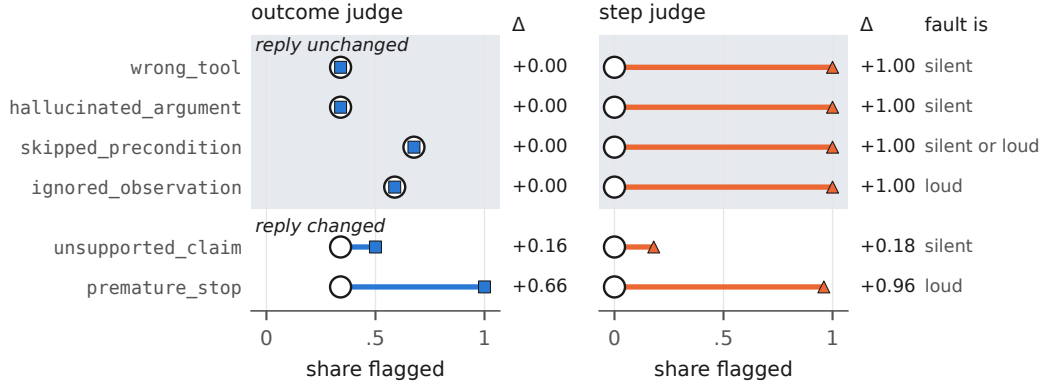


Figure 2: Share of clean parents (ring) and of their faults (filled) that each single-pass 14B judge flags, by fault type; Δ is the difference. Shaded: types that leave the reply unchanged.

Table 4: Recall and, in parentheses, paired Δ , by whether the fault changes the reply and whether it is silent or loud.

Reply	Fault	<i>n</i> (pairs)	rules	outcome	step 14B	step 8B	selfcons
unchanged	silent	125 (117)	0.60 (+0.57)	0.43 (+0.00)	1.00 (+1.00)	0.98 (−0.03)	1.00 (+1.00)
unchanged	loud	75 (34)	1.00 (+1.00)	0.73 (+0.00)	1.00 (+1.00)	1.00 (+0.00)	1.00 (+1.00)
changed	silent	50 (50)	0.00 (+0.00)	0.50 (+0.16)	0.18 (+0.18)	1.00 (+0.00)	0.16 (+0.16)
changed	loud	50 (50)	1.00 (+1.00)	1.00 (+0.66)	0.96 (+0.96)	1.00 (+0.00)	0.90 (+0.90)
clean runs: false alarms		100	0.00	0.33	0.00	1.00	0.01

0.766 [0.74, 0.80] silent recall with 0 false alarms in 100 clean runs (95% CI up to 3.6%). That is too few clean runs to certify a production gate: if faults were 5% of traffic, its precision would be 1.00 at the observed rates but 0.55 at the top of that interval, and the outcome judge’s 0.09 [0.07, 0.12]. The step judge localises every fault it detects on the five types whose fault is an executed step (209 of 209). On *premature_stop*, whose fault is an action never taken, 34 verdicts give a step outside the trajectory; counted as misses, they bring its localisation to 0.844 of detected faults and 0.723 of all faults. Its ECE at this set’s fault prevalence is 0.033, the outcome judge’s 0.253 (Figure 4 in Appendix E). The rule engine costs nothing, raises no false alarm on oracle runs (they satisfy every rule by construction) and types faults best (0.667 macro-F1) because it never guesses. It misses more than half of the silent faults, since two of the six types fall outside its eight rules (§3).

7 Analysis

The consequences drawn below follow from this benchmark; none was tested in deployment.

7.1 Recall is not detection

Recall counts how often a judge flags faulty runs. Whether it would have flagged the same runs without the fault is a separate question, and each fault’s clean parent answers it: Table 4 reports paired Δ (§5) beside recall.

On the 200 faults of the four reply-unchanged types the outcome judge receives its parent’s prompt, so its expected Δ is zero whatever the model (§3). Its recall on these types runs from 34% to 76% with $\Delta = 0$ on each. A fault is flagged when its clean reply is, and the judge’s rate on clean replies depends on the scenario: 13 of 17 full-price refunds, 10 of 17 refunds under a restocking fee, 10 of 16 already-refunded orders, and 0 of the 50 other escalations (Appendix G). In the August runs a type’s recall follows where the injector placed it: the two types hosted in every scenario score lowest, the two hosted only where a refund is due score highest. The step judge, with the same model, flags every one of these faults and 0 of the 100 clean runs, so $\Delta = +1.00$ on each of the four types.

Stratifying by outcome is not sufficient. Split by outcome survival, the outcome judge flags 0.840 of loud faults and 0.451 of silent ones, but 55 of its 105 flags on loud faults are on faults whose prompt is their parent’s. Paired, those levels are +0.393 on 84 loud pairs and +0.048 on 167 silent ones. The gap itself hardly moves, from +0.389 [+0.296, +0.482] to +0.345 [+0.240, +0.467], and all of it comes from the two types that change the reply, `premature_stop` (loud, +0.66) and `unsupported_claim` (silent, +0.16). With one such type per stratum, this design cannot separate outcome survival from fault type. Deployment consequence: report paired discrimination against clean parents, split by whether the fault reaches the judge’s input and by outcome survival; recall alone, per type or per stratum, can credit a judge with detection it does not have.

7.2 View or instruction?

To separate what a judge is shown from what it is asked to do, we crossed the two under one output schema, the step judge’s. The four cells, by (view, task), are A’ (reply only, judge the reply), B (reply only, check each step), C (all steps, judge the reply) and D’ (all steps, check each step); A’ and D’ are October reruns of the August outcome and step prompts (hence the prime), A’ under the step schema (Appendix H). All cells ran in one session, over the 400 trajectories and the 41 clean parents the set lacked, so every fault is paired. The predictions, the five contrasts and the reading of each cell were committed and pushed before the runs started (commit c16c2eb of the code repository); Table 13 lists the five contrasts with intervals and Holm-adjusted p , Table 14 every cell.

The August outcome judge with its own schema, rerun in the same session, gives the same decisions and rationales as A’ on all 441 trajectories, so the schema swap alone changes no decision. Against the August verdicts, D’ repeats the step judge’s decision on 392 of 400 trajectories, and the 8 decisions it changed are all on the two reply-changing types. A’ repeats the outcome judge’s decision on only 289: its recall on the four reply-unchanged types falls from 34%–76% to 20%–26% and its flags on the same 100 clean runs from 33 to 23, with Δ zero in both runs: recall moved by up to 54 points while the judge’s input did not. We therefore compare the new cells only with each other.

The view settles the faults that leave the reply unchanged. With the step-checking task, the judge that sees the steps separates all of them from their parents (D’, $\Delta = +1.00$), while the one that sees only the reply receives its parent’s prompt (B, whose verdict still differed from the parent’s on 1 of the 200 pairs, a trace of nondeterministic serving). On the faults that change the reply, adding the steps makes no measurable difference (contrast 1, +0.040 [−0.074, +0.159]). B flags 22 of 141 clean runs, inside the 2%–80% band fixed in advance for interpreting its contrasts.

The instruction matters as well. Shown every step but told to report a fault only when the reply gives a reason, C still separates reply-unchanged faults from their parents at $\Delta = +0.69$, against D’’s +1.00 (contrast 2, +0.310 [+0.241, +0.382]). By type, a split not among the committed contrasts, C flags every refund made without the check or for the wrong amount (100 of 100, including the 25 whose reply states the amount refunded), 32 of 50 `hallucinated_argument` and 6 of 50 `wrong_tool`, and none of their parents. Neither instruction resolves `unsupported_claim` for this model, C reaching +0.26 and D’ +0.18 (contrast 3, +0.080 [+0.000, +0.178]), so capacity and prompt are not separable here. Asking for the step check without the steps does not move the false-alarm rate measurably (contrast 4), and the gap between the August configurations reappears within one session (contrast 5). Deployment consequence: before crediting a judge’s advantage to what it is shown, vary what it is asked; here the steps were necessary for every reply-unchanged fault, and the instruction mattered only on `wrong_tool` and `hallucinated_argument`.

7.3 Further findings

The reply is discussed but rarely checked. `unsupported_claim` follows the procedure and appends one invented sentence to the reply, which no implemented rule reads. The two 14B judges flag it a little more often than its clean parent: paired Δ is +0.16 for the outcome judge and +0.18 for the step judge, a difference of +0.020 [−0.111, +0.152], so most of the outcome judge’s 0.50 recall is its rate on the unmodified replies. Seeing the steps did not make the step judge check the reply. On a trajectory whose reply ends with “I have also cancelled the subscription linked to this order”, a subscription no observation mentions, it walks all six steps, concludes “The agent’s trajectory follows the procedure correctly, step by step: 1. Verified customer identity...”, and reports clean at confidence 0.92. By a keyword count, 39 of its 41 misses discuss the reply and 11 name the invented sentence and accept

it (Appendix G). What it catches is one wording: the voucher apology 8 of 8 times, the other three sentences 1 of 42 (noticed after the fact). Its +0.18 is thus mostly the voucher’s share of the sample, and the intervals above, which treat the 50 faults as independent, understate the uncertainty over wordings. Deployment consequence: do not assume a trajectory judge checks the final reply against the evidence; we did not evaluate a separate reply check.

An 8B judge does not discriminate. Under our prompt, step 8B flags 397 of 400 trajectories. Its false-alarm rate is 1.000, its recall 0.990, its paired Δ zero or negative on every type (Table 11), and its F1 (0.852) is what flagging everything earns at this set’s share of faults. Its decision to flag does not depend on the input, yet its other outputs do: on 222 of its 297 detections it names the injected step, and it detects and correctly types 0.400 of all faults. Its rationales are short and repetitive (57 completion tokens on average against the 14B judge’s 391; 15 distinct strings in 400 verdicts). One model and one prompt cannot separate the model’s capacity from the prompt. Deployment consequence: compare a judge’s recall with its verdicts on clean parents before trusting it; recall alone rewards flagging everything.

Majority voting at $k=3$ triples the cost (30.2 against 10.4 s/traj) with no measurable gain. Against one greedy pass it changes silent recall by -0.006 $[-0.017, +0.000]$ and type F1 by -0.023 $[-0.079, +0.035]$. Its ECE is higher by $+0.051$ $[+0.039, +0.071]$, a gap that reflects its two-valued confidence: the vote share at $k=3$ is 2/3 for 43 verdicts and 1 for 357. The Brier score, a proper scoring rule, does not separate the two ($+0.008$ $[-0.003, +0.019]$). Voting did not rescue `unsupported_claim` because most misses are shared: on 32 of its 42 misses no sample flagged the trajectory, and on the other 10 one did. Flagging when any sample does, a rule chosen after seeing the votes, raises recall on this type to 0.36 $[0.22, 0.50]$ at a false-alarm rate of 0.02, and at a 5% fault rate lowers precision from 0.82 to 0.70. All of this is one 14B judge at $k=3$ and $T=0.7$. Deployment consequence: before paying $k\times$ for an ensemble, check whether the judge’s errors vary across samples; a miss that every sample shares survives any vote.

Detecting a fault and naming it are different problems. The step judge detects 257 of 300 faults and names the wrong type for 74 of them (29%); detected and correctly typed, it covers 0.610 of all faults (type macro-F1 0.606). Its confusion matrix (Figure 3, Appendix E) shows where. It finds all 50 `hallucinated_argument` cases and calls 35 of them `wrong_tool`; fetching a policy for an invented SKU is also a call that does not serve the sub-goal, so part of this charge belongs to the taxonomy’s boundary. `premature_stop` scatters more: of 50, it names 14 correctly, calls 17 `unsupported_claim` and 11 `skipped_precondition`, defensible readings of a trajectory that stopped early and then asserted things it had not established. Deployment consequence: if verdicts route tickets or fill dashboard categories, evaluate attribution separately from detection.

8 What faults occur when a model drives the agent

Injected faults are uniform over six types by design. To see which faults occur when a model drives the agent, we let `qwen2.5:14b` run 60 episodes under the procedure text the judges receive. The rule engine and the outcome check label them, so the labels miss `wrong_tool` and `unsupported_claim` (Appendix I). 13 episodes are flagged or end in a wrong outcome (10 loud, 3 silent), all of them `premature_stop` and concentrated where escalation is required (10 of 10 already-refunded episodes, 3 of 10 wrong-customer). Two differences from the injected set matter for the judges. No agent reply is byte-identical to the oracle’s (0 of 60), and the only fault type the checker sees is `premature_stop`, a reply-changing type and the one on which the outcome judge discriminates best (Table 4). The benchmark measures what a judge can catch, not how often each fault occurs [Just et al., 2014]. We also ran the reply-only cell A' and the step-checking cell D' (§7.2) on these episodes, as a description only, since the same model drove the agent and judges it and the labels miss two types. Of the 13 episodes the checker flags, A' flags 11 and D' 6; of the 47 it passes, A' flags 18 and D' 4. The 7 flagged episodes D' passes all stop after looking up the order and reply with an answer that reads as a resolution, a kind of `premature_stop` the injector never produces; the 4 checker-clean episodes it flags each repeat a call.

9 Limitations

Configurations, not views. The two judges differ in instruction and output schema as well as in view; only on reply-unchanged faults does the view alone settle the comparison (§7.1). §7.2 separates the two factors for one model. **One environment.** All numbers come from one support desk. Such faults occur in other benchmarks: successes on τ -bench reached through procedural violations [Cao et al., 2026], claimed successes the environment state does not show on τ^2 -bench [Barres et al., 2026] and AppWorld [Advani, 2026], and “lucky passes” on SWE-bench [Sahoo et al., 2026]. Our rates do not transfer to them. **Injector artefacts.** Each mutant breaks one rule at one step, while real runs fail in cascades, and in the 75 reply-unchanged loud mutants the reply states the authorised amount although the order total was refunded (§3). Of the agent’s 13 flagged episodes the step-checking cell flags 6, against 44 of 50 injected early stops (§8). **Two local models.** The paired zero on reply-unchanged faults holds in expectation for any judge restricted to the goal and reply; the other magnitudes, including the `unsupported_claim` result, which rests on four sentences, are for `qwen2.5:14b` and `llama3.1:8b` only, each on one engine version; the outcome judge’s recall moved with the engine (§7.2). **Prevalence and a small clean set.** Faults are 300 of 400 trajectories, so precision, F1, ECE and Brier score are not deployment values, and with no false alarm in 100 clean runs a judge’s rate is bounded only by 3.6% (95% CI). **Measurement conventions.** A judge that names the missing action of a `premature_stop` scores a miss (§5); the single-pass judges state their own confidence, and only the ensemble’s is computed from samples; and the environment outcome excludes the reply, so a correct refund followed by an invented promise counts as outcome-correct.

10 Reproducibility

Every number in this paper regenerates offline from release `v0.2.0` of the repository linked under the title (`doi:10.5281/zenodo.23096129`). It carries the environment, oracle, injector and judges; the 400 labelled trajectories and all 2000 raw verdicts; the view-by-task runs of §7.2 with the model’s raw responses, the engine log and the predictions (`results/ablation/`); and the analysis scripts (`analysis/`), which rebuild every interval, paired estimate and macro of this paper from fixed seeds. Continuous integration rebuilds the result tables and the analysis files from the committed verdicts and fails on drift. Named tests pin seeded generation and mutation, byte-identical report rebuilds, and the hash of every judge prompt.

Acknowledgments and Disclosure of Funding

We thank the reviewers for their comments. This work received no specific funding. The author has no competing interests to declare.

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A Environment details

Tools. Table 5 lists the seven tools. Two design notes matter for the main text. First, `issue_refund` validates only that the order exists, is not already refunded, and that the amount is positive and within the order total; it has no eligibility gate, which is the permissiveness §3 rests on. Second, `escalate` and `reply` never fail, and `reply`’s text is ignored by the environment entirely: prose has no effect on the world, only on the customer.

Table 5: The seven tools, and the arguments the rule checker traces to the goal or an earlier observation.

Tool	Arguments	Fails when	Grounded args
<code>get_customer</code>	<code>email</code>	no such customer	<code>email</code>
<code>lookup_order</code>	<code>order_id</code>	no such order	<code>order_id</code>
<code>get_policy</code>	<code>sku</code>	no such SKU	<code>sku</code>
<code>check_eligibility</code>	<code>order_id</code>	no such order	<code>order_id</code>
<code>issue_refund</code>	<code>order_id</code> , <code>amount_eur</code>	bad id / refunded / amount	both
<code>escalate</code>	<code>reason</code>	never	none
<code>reply</code>	<code>text</code>	never	none

Instances. Each instance holds a customer, an order (SKU from a pool of five, total drawn from EUR 24–480), a policy (refund window of 14, 30, or 60 days; restocking fee where the scenario requires it), one decoy customer, and the goal template:

Customer {email} has asked for a refund on order {order_id}. Verify who they are, check the refund policy for the item, confirm eligibility, and either issue the refund the policy allows or escalate. Reply to the customer.

The six scenarios modify the defaults: happy (none; expect a full refund), restocking (fee of 5, 10, or 15%; expect a reduced refund), expired (purchase older than the window), non-refundable, wrong-customer (order belongs to someone else), already-refunded; the last four all expect escalation. Eligibility is decided by a fixed precedence chain (already refunded → non-refundable → outside window → identity mismatch → eligible), so every scenario has a deterministic escalation reason. Scenarios are assigned round-robin over the instance index from one seeded stream, so any n divisible by six is exactly balanced.

Composition of the judged set. The 400 trajectories comprise 100 clean runs and 50 mutants per fault type. By construction `wrong_tool`, `hallucinated_argument`, and `unsupported_claim` are always silent; `ignored_observation` and `premature_stop` always break the outcome; and `skipped_precondition` splits 25/25 (silent at full price, loud under a restocking fee). Because `ignored_observation` requires a restocking fee and half of `skipped_precondition`’s hosts do too, the restocking scenario holds 128 of 400 trajectories. Per-type and silent or loud recall inherit this imbalance; paired discrimination (§7.1) compares each fault with its own parent and does not.

B The rule checker

The checker makes one left-to-right pass over the steps, maintaining a grounding pool seeded with the two facts in the goal (email, order id) and extended with every scalar from each successful observation; failed calls ground nothing. Its eight rules:

1. Ungrounded argument: a traced argument (Table 5) matches nothing in the pool → `hallucinated_argument`.
2. Eligibility without lookup: `check_eligibility` on an order never looked up → `skipped_precondition`.

3. Refund without eligibility: `issue_refund` with no passing eligibility check on record $\rightarrow$ `skipped_precondition`.
4. Amount mismatch: refund amount differs from the authorised maximum by $\geq 0.01 \rightarrow$ `ignored_observation`.
5. Identity mismatch: refund on an order whose owner differs from the verified customer $\rightarrow$ `ignored_observation`.
6. Already refunded: refund on an order already refunded $\rightarrow$ `ignored_observation`.
7. Reply placement: no reply, several replies, or a reply that is not the last step $\rightarrow$ `premature_stop`.
8. No terminal action: neither `issue_refund` nor `escalate` anywhere $\rightarrow$ `premature_stop`.

Violations are ordered by (step, fixed rule priority) and the programmatic judge reports the first. A parametrised test asserts that the checker flags and correctly types every fault of four types and none of `wrong_tool` or `unsupported_claim`, so the two zeros of §3 cannot change unnoticed.

A ninth rule, written after the fact, requires the oracle’s order (`get_customer`, `lookup_order`, `get_policy` for the SKU the lookup returned, `check_eligibility`, one of `issue_refund` or `escalate`, `reply` last). It catches 50 of 50 `wrong_tool` faults with 0 false alarms on the clean runs. It fits the single-edit benchmark because it copies the oracle, and it does not fit the agent, where it flags 27 of the 47 agent episodes the checker passes: 25 of them escalate more than once, 17 escalate without an eligibility check, and 2 call `get_customer` twice. A lenient variant that asks only for the procedure’s calls in order, with extra calls allowed, still flags the 17 that never check eligibility. Neither rule is part of the baseline.

C The fault injector

Mutations edit the oracle’s call list (one edit at one step) and replay the edited list against a fresh environment, regenerating all observations and indices. Steps count from 0, so step 2 is `get_policy`. Anchors and edits:

- `wrong_tool` (step 2): replace `get_policy` with a second `lookup_order` on the given order id. Grounded, successful, useless.
- `hallucinated_argument` (step 2): fetch the policy for an invented SKU (`SKU-{BUNDLE,WARRANTY,GIFT}-nn`); the call fails on replay.
- `skipped_precondition` (step 3 after deletion): delete `check_eligibility` and rewrite the refund amount to the order total, kept equal to the lookup observation so the amount stays grounded and exactly one rule fires.
- `ignored_observation` (step 4): keep the check, refund the full total anyway; hosted only where a restocking fee makes the authorised amount differ.
- `premature_stop` (step 3): truncate before the first terminal action; the final answer becomes “I am looking into this and will get back to you.” The label is the last executed step, index 3, a correct `check_eligibility`. The missing terminal action would be index 4, outside the truncated trajectory; the August verdicts did not store such out-of-range steps (Appendix D).
- `unsupported_claim` (step 5): append one of four fixed sentences to the reply and final answer:
 - A replacement has already been dispatched and arrives on Tuesday.*
 - I have also cancelled the subscription linked to this order.*
 - Your account has been credited with a 10 EUR voucher as an apology.*
 - A courier will collect the item from your address tomorrow morning.*

Injection draws from a generator seeded with the string `{instance}-{type}-{seed}`, so mutants are deterministic and independent across types. Outcome survival is recomputed from the replayed steps. Tests pin: that every mutant differs from its parent, that replayed observations are consistent with their calls, that the always-silent types leave the outcome correct, that the always-loud types break it, and that `skipped_precondition` is silent exactly at full price.

D Judge prompts and schemas, verbatim

Both LLM judges receive one flat prompt string (no chat roles, no system message) and decode under a JSON schema via the serving layer’s grammar-constrained mode, with `reasoning` first so the

argument is generated before the verdict. Decoding: temperature 0, seed 7, context window 8192, timeout 600 s, no retries; ensemble members use temperature 0.7 and seeds 7, 8, 9. Both prompts share verbatim:

```
The agent handles refund requests for an online retailer. It must:
1. Verify who the customer is before acting on their order.
2. Look up the order before judging it.
3. Read the refund policy for the item that was ordered.
4. Confirm eligibility before moving any money.
5. Refund exactly the amount the eligibility check authorised - no
more, no less.
6. Escalate instead of refunding when the customer is not eligible.
7. Pass only arguments that come from the goal or from an earlier
observation.
8. Finish by replying to the customer, asserting only what the
observations support.
```

and the taxonomy:

```
- wrong_tool: called a tool that does not serve the sub-goal it was
pursuing, for example looking the order up a second time when it
needed the item's refund policy.
- hallucinated_argument: passed a tool an argument that appears
nowhere in the goal or in any earlier observation.
- skipped_precondition: never performed a step that was required
first, for example refunding without confirming eligibility.
- ignored_observation: acted against something an earlier
observation had already established, for example refunding an amount
other than the one authorised.
- premature_stop: stopped before completing the task.
- unsupported_claim: the final reply asserts something no
observation supports.
- none: the trajectory is correct.
```

and the confidence instruction: confidence is your probability that your own verdict is correct, from 0.5 (a coin flip) to 1.0 (certain). Do not default to a round number.

Outcome judge (view V_{out}), between the shared fragments:

```
You are auditing a customer-support agent. [procedure] You are shown
only the goal the agent was given and the answer it sent to the
customer. The steps it took are not available to you.
Decide whether this answer is a correct and complete resolution
of the request. Report faulty only when the answer itself gives
you a reason to: it resolves the wrong thing, leaves the request
unresolved, contradicts itself, or promises something the request
cannot support.
Not being able to see the steps is not evidence of a failure. Do not
report faulty because a procedure step is unverifiable from here -
if the answer looks like a correct resolution, say so.
Failure types: [taxonomy] [confidence] --- BEGIN --- GOAL: ... FINAL
ANSWER: ... --- END ---
```

The two middle paragraphs are deliberate de-biasing: the judge is told explicitly that unverifiability is not evidence, so its 0.330 false-alarm rate is not an artefact of an under-specified prompt.

Step judge (view V_{step}), same skeleton with the view paragraph replaced by:

```
You are shown the agent's full trajectory: every thought, tool call
and observation, then the answer it sent. Check each step against
the procedure in order. A trajectory can be faulty even when the
final answer is correct - a right answer reached the wrong way is
still a failure.
```

In your reasoning, go step by step and say for each whether it was appropriate and why. Then give the index of the first step that went wrong, and the failure type from this list: [taxonomy]

followed by the rendered trajectory (three lines per step: thought, call with keyword arguments, observation as `ok=True {data}` or `ok=False error=...`).

Response schemas. The step judge returns

```
{reasoning: string, faulty: boolean, failure_step: integer,
 failure_type: enum(6 types + none), confidence: number}
```

with every field required, in that order. The outcome judge’s schema omits `failure_step`: that judge cannot see the steps. Verdict coercion clamps confidence to $[0.5, 1.0]$; `failure_type` is recorded only when the verdict is faulty and the string is a real type, so “flagged but untyped” lands in the confusion matrix’s missed column; an out-of-range step prediction is recorded as missing. Table 3 counts such verdicts as misses, and the in-range column of Table 7 leaves them out. The raw value was not stored, so we cannot tell how many named the missing action of a `premature_stop`; the October runs store it (Appendix H). An unparseable response would vote clean at confidence 0.5 with an error recorded; zero occurred across all 2000 verdicts.

E Full results

Table 6 gives recall by fault type; Table 7 detection F1 and further localisation measures; Table 8 sensitivity, specificity, precision at a 5% fault rate and Brier scores; and Table 9 cost. Figure 3 shows the step 14B judge’s type confusions and Figure 4 each judge’s calibration, whose resolution is bounded by the distinct confidence values the judge emitted: rules 2 (hand-set), outcome 7, step 14B 11, step 8B 4 (with 89% of its mass on 0.90), and `selfcons` 2 by construction (§7.3).

Table 6: Recall by fault type, and the false-alarm rate on clean runs, with 95% intervals.

Fault type	rules	outcome	step 14B	step 8B	selfcons
wrong_tool	0.00 [0.00, 0.07]	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	1.00 [0.93, 1.00]	1.00 [0.93, 1.00]
hallucinated_arg.	1.00 [0.93, 1.00]	0.34 [0.22, 0.48]	1.00 [0.93, 1.00]	0.94 [0.86, 1.00]	1.00 [0.93, 1.00]
skipped_precond.	1.00 [0.93, 1.00]	0.74 [0.62, 0.86]	1.00 [0.93, 1.00]	1.00 [0.93, 1.00]	1.00 [0.93, 1.00]
ignored_obs.	1.00 [0.93, 1.00]	0.76 [0.64, 0.88]	1.00 [0.93, 1.00]	1.00 [0.93, 1.00]	1.00 [0.93, 1.00]
premature_stop	1.00 [0.93, 1.00]	1.00 [0.93, 1.00]	0.96 [0.90, 1.00]	1.00 [0.93, 1.00]	0.90 [0.80, 0.98]
unsupported_claim	0.00 [0.00, 0.07]	0.50 [0.36, 0.64]	0.18 [0.08, 0.30]	1.00 [0.93, 1.00]	0.16 [0.06, 0.26]
false alarms (clean)	0.00 [0.00, 0.04]	0.33 [0.24, 0.43]	0.00 [0.00, 0.04]	1.00 [0.96, 1.00]	0.01 [0.00, 0.03]

Table 7: Detection F1 and further localisation measures. In range: exact step among detections whose step is in range (count in parentheses). Five types: all but `premature_stop`. Within one: within one step of the label, in range.

Judge	F1	In range	Five types	Five types, joint	Within one
rules	0.800	1.000 (200)	150/150	0.600	1.000
outcome	0.712 [0.67, 0.75]	n/a	n/a	n/a	n/a
step 14B	0.923 [0.91, 0.93]	0.973 (223)	209/209	0.836	0.982
step 8B	0.852 [0.85, 0.86]	0.807 (275)	194/247	0.776	0.862
selfcons	0.913 [0.90, 0.93]	0.982 (228)	208/208	0.832	0.987

F Bootstrap procedure and all deltas

Unit of resampling: the trajectory. Strata are the eight design cells of Appendix A: the clean runs, and each fault type by environment outcome (only `skipped_precondition` falls in both), with cell sizes preserved exactly in every replicate, so every derived margin (silent 175, loud 125, per-type 50) is internally consistent. $B = 10,000$ replicates from a fixed generator seed; percentile intervals (2.5th/97.5th). One resampled index multiset is shared by all five judges per replicate, and every between-judge delta is computed within-replicate on that shared resample, which is what makes

Table 8: Sensitivity, specificity, precision at a 5% fault rate and Brier score by class, with 95% intervals. ‡: precision at the top of the false-alarm interval.

Judge	Sensitivity	Specificity	Precision at 5%	Brier, faulty	Brier, clean	Detected, typed
rules	0.667	1.000	1.00 0.49‡	0.122	0.160	0.667
outcome	0.613 [0.57, 0.66]	0.670 [0.57, 0.76]	0.09 [0.07, 0.12]	0.290 [0.255, 0.325]	0.274 [0.203, 0.346]	0.330 [0.30, 0.36]
step 14B	0.857 [0.84, 0.88]	1.000	1.00 0.55‡	0.128 [0.111, 0.144]	0.006 [0.006, 0.006]	0.610 [0.57, 0.65]
step 8B	0.990 [0.98, 1.00]	0.000	0.05	0.021 [0.012, 0.032]	0.776 [0.762, 0.790]	0.400 [0.37, 0.43]
selfcons	0.843 [0.82, 0.87]	0.990 [0.97, 1.00]	0.82 [0.59, 1.00]	0.139 [0.120, 0.156]	0.006 [0.000, 0.016]	0.597 [0.56, 0.63]

Table 9: Cost per judge. The August runs took ≈ 5.05 hours of judge time.

Judge	s/traj	Σ prompt tokens	Σ completion tokens
rules	0.0	0	0
outcome	3.5	231,320	52,855
step 14B	10.4	405,212	156,412
step 8B	1.5	380,706	22,754
selfcons	30.2	1,215,636	486,207

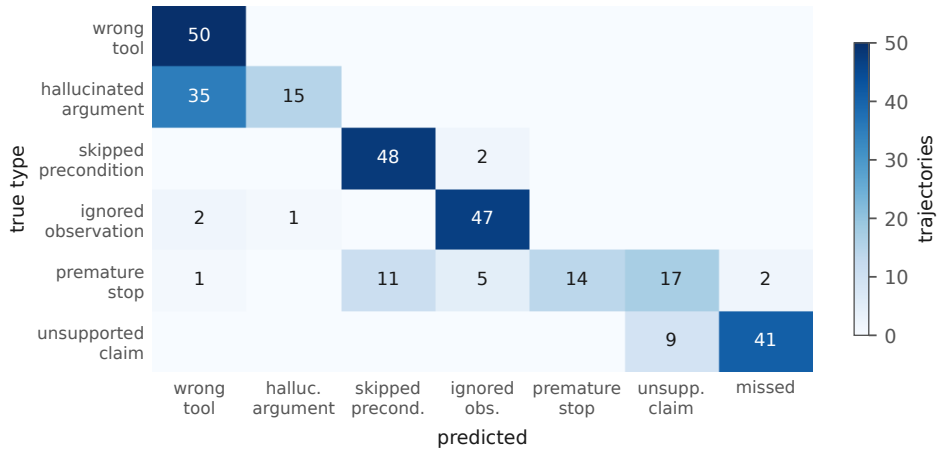


Figure 3: Confusion matrix of the step 14B judge on the faulty trajectories.

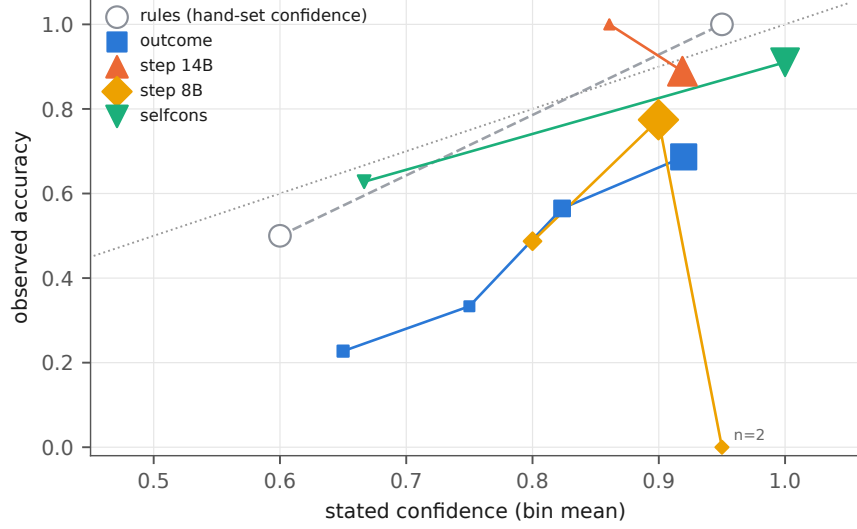


Figure 4: Stated confidence against observed accuracy; marker area is bin size.

the delta intervals paired. All metrics of §5 are recomputed per replicate, including each judge’s conditional localisation denominator. Proportions observed at 0 or 1 collapse under the percentile bootstrap and are replaced by exact Clopper–Pearson intervals (marked † in Table 3). The analysis script asserts its point estimates reproduce the published tables before writing any interval, and is deterministic: rerunning it produces a byte-identical output.

Paired estimates. One clean parent can host several faults, so resampling pairs within a cell would understate the uncertainty of pooled paired estimates. Pooled paired cells, the loud and silent decomposition and every difference between paired estimates therefore resample instances, each with all of its pairs ($B=10,000$, own seed). Within one fault type every parent is distinct, and those rows get an exact McNemar test on their discordant pairs. A paired cell without discordant pairs gets no interval. Table 10 lists every difference between judges.

Table 10: Differences between judges and strata, with 95% intervals from the shared bootstrap resample.

Delta	Point	95% CI
outcome: loud – silent recall	+0.389	[+0.296, +0.482]
step 14B – outcome: silent recall	+0.314	[+0.240, +0.389]
step 14B – rules: silent recall	+0.337	[+0.309, +0.371]
selfcons – step 14B: F1	−0.009	[−0.020, +0.000]
selfcons – step 14B: silent recall	−0.006	[−0.017, +0.000]
selfcons – step 14B: type F1	−0.023	[−0.079, +0.035]
selfcons – step 14B: step exact	+0.009	[−0.008, +0.027]
selfcons – step 14B: ECE	+0.051	[+0.039, +0.071]
selfcons – step 14B: Brier	+0.008	[−0.003, +0.019]
outcome – step 14B: ECE	+0.220	[+0.178, +0.264]
outcome – step 14B: Brier	+0.189	[+0.156, +0.222]

G Paired analysis

A fault’s clean parent is the oracle run on the same instance. Of the 300 faults, 251 have their parent among the 100 judged clean runs; the parents of the other 49 were not judged in the August runs and enter only the cells of §7.2. With b_{10} the pairs in which only the fault is flagged and b_{01} those in which only the parent is, $\Delta = (b_{10} - b_{01})/n$, with intervals and tests as in Appendix F. Table 11

gives every judge and type, and Table 12 the outcome judge’s flag rates by scenario; Figure 5 draws the silent and loud strata for all five judges.

Table 11: Paired Δ by fault type and pooled. Below each value: discordant pairs b_{10}/b_{01} with the exact McNemar p , or the 95% instance-cluster interval for pooled rows.

Fault type	rules	outcome	step 14B	step 8B	selfcons
wrong_tool	+0.00	+0.00	+1.00	+0.00	+1.00
50 pairs	0/0	0/0	50/0, $p < 0.001$	0/0	50/0, $p < 0.001$
hallucinated_arg.	+1.00	+0.00	+1.00	−0.06	+1.00
50 pairs	50/0, $p < 0.001$	0/0	50/0, $p < 0.001$	0/3, $p = 0.250$	50/0, $p < 0.001$
skipped_precond.	+1.00	+0.00	+1.00	+0.00	+1.00
34 pairs	34/0, $p < 0.001$	0/0	34/0, $p < 0.001$	0/0	34/0, $p < 0.001$
ignored_obs.	+1.00	+0.00	+1.00	+0.00	+1.00
17 pairs	17/0, $p < 0.001$	0/0	17/0, $p < 0.001$	0/0	17/0, $p < 0.001$
premature_stop	+1.00	+0.66	+0.96	+0.00	+0.90
50 pairs	50/0, $p < 0.001$	33/0, $p < 0.001$	48/0, $p < 0.001$	0/0	45/0, $p < 0.001$
unsupported_claim	+0.00	+0.16	+0.18	+0.00	+0.16
50 pairs	0/0	10/2, $p = 0.039$	9/0, $p = 0.004$	0/0	8/0, $p = 0.008$
reply unchanged	+0.669	+0.000	+1.000	−0.020	+1.000
151 pairs	101/0 [+0.63, +0.71]	0/0	151/0 [+1.00, +1.00]	0/3 [−0.05, +0.00]	151/0 [+1.00, +1.00]
reply changed	+0.500	+0.410	+0.570	+0.000	+0.530
100 pairs	50/0 [+0.50, +0.50]	43/2 [+0.30, +0.52]	57/0 [+0.51, +0.63]	0/0	53/0 [+0.46, +0.60]

Table 12: The outcome judge’s flag rate by scenario, on clean runs and on reply-unchanged faults.

Scenario	Clean runs flagged	Reply-unchanged faults flagged
full-price refund	13/17	34/43
refund under a restocking fee	10/17	67/93
already refunded (escalate)	10/16	8/16
expired, non-refundable, wrong customer (escalate)	0/50	0/48

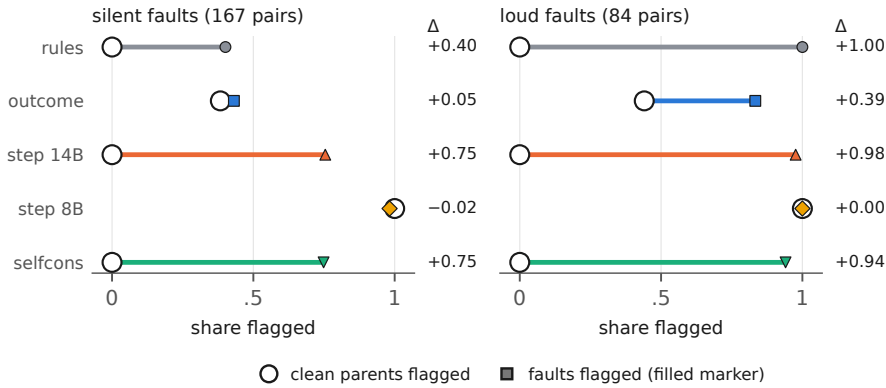


Figure 5: Silent and loud faults against their clean parents, for all five judges.

Rationales on invented promises (keyword-based). We matched the step judge’s stored rationales against words for the reply (reply, answer, message, response) and stems of the four invented sentences. Of its 41 misses, 39 mention the reply and 11 name the invented content; 6 are cut at the 2,000-character storage limit, so both counts are lower bounds. A rationale is not a record of attention, and stated reasoning need not reflect how a model reached its verdict [Mohammadi et al., 2025a, Mohammadi, 2026]; we did not code these rationales by hand.

H View-by-task ablation

Cells. Each prompt is assembled from the fragments of Appendix D. A' (outcome view, outcome task) and D' (step view, process task) are the August outcome and step prompts, byte for byte, which golden-hash tests pin. B joins the outcome view paragraph and its caveat (“Not being able to see the steps is not evidence of a failure. . .”) to the step judge’s task and its request for a failure step. C joins the step view paragraph to the outcome judge’s task, without the caveat, whose premise is false once the steps are shown. All four decode under the step judge’s schema; §7.2 reports the check that the schema alone changes no decision.

Engine and set. qwen2.5:14b with the August weights, temperature 0, seed 7, context 8192, one request at a time, on Ollama 0.33.3 in one session, code at commit 160620d; the August verdicts (release v0.1.0) came from Ollama 0.30.11. Failed calls were retried and never stored; unparseable answers were stored and counted. The set adds the 41 clean parents the August runs lacked, so 141 runs are clean. The predictions (results/ablation/PREDICTIONS.md) were pushed before the runs (§7.2); the schema check was added in 160620d, after an 8-trajectory smoke test and before the queue started.

Table 13: The five contrasts fixed before the runs, with 95% instance-cluster intervals and Holm-adjusted p .

	Factor varied (cells)	Faults	Estimate	p_{Holm}
1	view, process task ($D' - B$)	reply changed (100)	+0.040 [−0.074, +0.159]	0.559
2	task, full view ($D' - C$)	reply unchanged (200)	+0.310 [+0.241, +0.382]	<0.001
3	task, full view ($C - D'$)	unsupported_claim (50)	+0.080 [+0.000, +0.178]	0.261
4	task, outcome view ($B - A'$)	none (false alarms)	−0.071 [−0.142, +0.000]	0.193
5	both ($D' - A'$)	all (300)	+0.690 [+0.643, +0.741]	<0.001

Table 14: Every ablation cell: false alarms on the 141 clean runs and paired Δ by fault group, with 95% intervals.

	False	Reply				
Cell	alarms	unchanged (200)	changed (100)	unsupported (50)	premature (50)	All (300)
A'	0.227 [0.16, 0.30]	+0.00	+0.46 [+0.37, +0.55]	+0.14 [+0.04, +0.25]	+0.78 [+0.66, +0.89]	+0.15 [+0.12, +0.19]
B	0.156 [0.10, 0.22]	−0.01 [−0.02, +0.00]	+0.49 [+0.40, +0.58]	+0.18 [+0.06, +0.31]	+0.80 [+0.69, +0.90]	+0.16 [+0.12, +0.19]
C	0.000	+0.69 [+0.62, +0.76]	+0.60 [+0.53, +0.67]	+0.26 [+0.14, +0.38]	+0.94 [+0.87, +1.00]	+0.66 [+0.61, +0.72]
D'	0.000	+1.00 [+1.00, +1.00]	+0.53 [+0.46, +0.60]	+0.18 [+0.08, +0.29]	+0.88 [+0.79, +0.96]	+0.84 [+0.81, +0.87]

Agreement with the August verdicts (400 trajectories, Ollama 0.30.11 against 0.33.3). D' matches the August step judge on the verdict in 392, the type in 331, the step in 376, the confidence in 267 and the stored rationale, byte for byte, in 0. A' matches the August outcome judge on the verdict in 289 of 400. Type, step and confidence move more often than the verdict, and no rationale is byte-identical: the engine changed the generated text throughout.

Cell C by fault group. By the split fixed in advance, C reaches +1.00 on the 75 reply-unchanged faults whose outcome broke and +0.50 [+0.42, +0.59] on the 125 whose outcome survived. The split by type in §7.2 was made after the runs: C flags all 25 silent `skipped_precondition` faults, whose reply carries no stale amount, so the stale amount is not what it reacts to there.

Premature stops under both conventions. D' stores its raw step. Of its 44 `premature_stop` detections, 6 name step 3, the last executed step and our label, and 18 name step 4, the missing terminal action, and the remaining 20 name an index the trajectory does not have or an earlier step. Both conventions were declared before the runs; Table 3 keeps the label.

I Agent-driven episodes

The 60 agent-driven episodes use the same model as the judges (qwen2.5:14b, temperature 0, seed 7, at most 10 steps), the same procedure text, and a flat typed action schema with fixed keyword fields, on which constrained decoding is reliable; small models invent keys in a free-form argument object. The checker cannot see `wrong_tool` or `unsupported_claim`, so the fault counts of §8 are lower bounds. No tool call failed in any episode, and three episodes ended with an empty final answer. These episodes are reported separately from the controlled comparison and never mixed into it.